Final Report

STAT 7100 Project

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Overview

The initial premise of this project was initially kept very simple. Due to factors of this being the first statistical analysis I have ever undertaken and also, the focus was less on reaching a statistical conclusion, but rather implementing tests and models learned in STAT 7100. However, on preliminary analysis, there were numerous issues relating to the aviation dataset. By nature, aviation dataset can be considered noisy because of 10+ factors affecting the (arrival delay) dependent variable. During my preliminary analysis I also discovered that due to policy changes on how flight data is reported in 2018, I had to omit all years prior from the sample. Analysis was also shortened from comparing 51 reporting airlines to only doing the 10 marketing carriers they fall under, to avoid making the project too big.

There are several limitations with this project currently. Delay is a variable affected by dozens of factors, all of which are not even addressed in this project or even present in the dataset itself.

In aviation, as an industry standard a flight landing within 15 minutes is considered on-time, hence the presence of the arr_del15 variable in the dataset. However, for this project, I am considering any deviation from the scheduled arrival time to be a delay, hence we will be using the pure arrival delay (in minutes) as the main factor for this project.

By comparison on the mean delay at airports where all carriers were present, using concepts such as ANOVA and Bootstrapping, I concluded that, Alaska Airlines is the best performing when it comes to delays if we look at the statistics only

However, the season trends showed that this is due to Alaska not flying the same congested summer routes as other airlines, which brings into contention the next closest runner up, Southwest Airlines. The trend was highlighted during the second part of the project which was the Holt-Winters time series forecast, that showed that Southwest performed close to Alaska Airines while also facing a severe spike in delays during the summer. Only trailing by 1.3 minutes.

Introduction

EDA (Preliminary Analysis):

The dataset used for this project is a record for U.S. airlines on-time performance, which measures flight delays in minutes. It is taken from Kaggle but the original sourced is the Bureau of Transportation Statistics, it tracks monthly performance for every major carrier going back to 2003 up till the beginning of 2025.

Due to the origin of this dataset, and the fact that it had been already checked by the user that uploaded it on Kaggle, there was not a requirement to conduct any preliminary analysis as dataset was valid. Using my own background in aviation I was also able to check running basic checks on the Mean, SD, Variance and Max/Min values for all variables to ensure that the dataset is suitable for this project.

Key Variables:

marketing_carrier: The primary grouping variable for the analysis. This identifies the airline brand selling the ticket regardless of which regional partner actually operated the flight. This variable did not come with the dataset, it is a new grouping based on carrier_name (name of the reporting airline) and carrier (airline code) variables

airport: The three-letter IATA code (eg. JFK, ATL, EWR) identifying the airport. This was important for the "Common Airport" analysis to check for common airports for all 10 carriers

year & month: Used to construct the timeline from 2018–2025 for the trend analysis and the Holt-Winters forecasting model

Data Cleaning Steps

As previously mentioned, data was already suitable for this project and required very little cleaning steps.

First, I grouped 51 reporting airlines under their 10 main marketing carriers (like moving Delta Connection flights under Delta) so the totals were correct. I also fixed inconsistent airport names to make sure every airport code matched exactly one name. To prevent errors in my math, I removed impossible negative delay numbers and fixed rows that listed zero flights.

```r
airdata_raw <- read.csv("Airline_Delay_Cause.csv", stringsAsFactors = FALSE)
test <- sum(airdata_raw$arr_flights, na.rm = TRUE)

dim(airdata_raw) # 398233 obs. of  21 variables
str(airdata_raw)
head(airdata_raw)
summary(airdata_raw)
airdata_raw <- na.omit(airdata_raw)
sd(airdata$arr_del15)
sd(airdata$arr_delay)
mean(airdata$arr_delay)
mean(airdata$arr_del15)




max_delay_row <- airdata_raw[which.max(airdata_raw$arr_delay), ]
max_delay_row

# Airline and Carrier Code Count

table_carrier <- table(airdata_raw$carrier)
table_carriername <- table(airdata_raw$carrier_name)
length(table_carrier)
length(table_carriername)




# Total Flights per carrier
flight_totals <- aggregate(arr_flights ~ carrier_name, data = airdata_raw, FUN = sum)
flight_totals_sorted <- flight_totals[order(flight_totals$arr_flights, decreasing = TRU
print(flight_totals_sorted)


#Airline Service Span
min_years <- aggregate(year ~ carrier_name, data = airdata_raw, FUN = min)

max_years <- aggregate(year ~ carrier_name, data = airdata_raw, FUN = max)
```

Example of some of the preliminary analysis carried out on the CSV table taken from Kaggle.com

The span of data was truncated from the default 2003-2025, to 2018-2025, this was done for two reasons. Policy changes in 2018 affected the way airlines reported their flight statistics and to account for all reporting airlines being active in the sample time frame.

Models and Visualisation

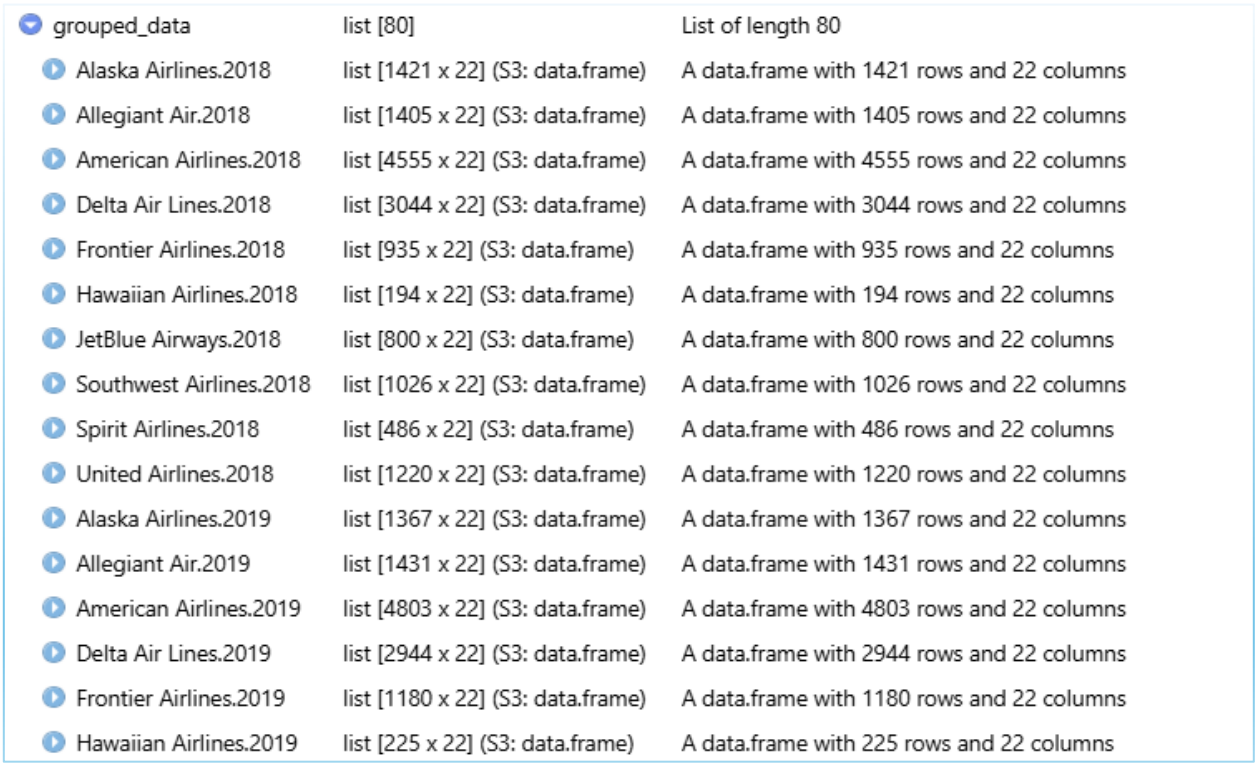
For this project, 3 statistical concepts were employed, these are:

1. Confidence Intervals generated by Non-Parametric Bootstrapping

This was applied for the average yearly delay chart. The reason for choosing this method is because airline on-time performance datasets and most real world data is not actually conforming to distributions like Normal or Poisson. Using parametric would be to assume that our dataset was normally distributed, which was not the case.

This model was applied by using subset() function in R to create individual dataset for each marketing for each year

(10 Marketing Carrier x 8 Years = 80 Datasets)



grouped_data	list [80]	List of length 80
Alaska Airlines.2018	list [1421 x 22] (S3: data.frame)	A data.frame with 1421 rows and 22 columns
Allegiant Air.2018	list [1405 x 22] (S3: data.frame)	A data.frame with 1405 rows and 22 columns
American Airlines.2018	list [4555 x 22] (S3: data.frame)	A data.frame with 4555 rows and 22 columns
Delta Air Lines.2018	list [3044 x 22] (S3: data.frame)	A data.frame with 3044 rows and 22 columns
Frontier Airlines.2018	list [935 x 22] (S3: data.frame)	A data.frame with 935 rows and 22 columns
Hawaiian Airlines.2018	list [194 x 22] (S3: data.frame)	A data.frame with 194 rows and 22 columns
JetBlue Airways.2018	list [800 x 22] (S3: data.frame)	A data.frame with 800 rows and 22 columns
Southwest Airlines.2018	list [1026 x 22] (S3: data.frame)	A data.frame with 1026 rows and 22 columns
Spirit Airlines.2018	list [486 x 22] (S3: data.frame)	A data.frame with 486 rows and 22 columns
United Airlines.2018	list [1220 x 22] (S3: data.frame)	A data.frame with 1220 rows and 22 columns
Alaska Airlines.2019	list [1367 x 22] (S3: data.frame)	A data.frame with 1367 rows and 22 columns
Allegiant Air.2019	list [1431 x 22] (S3: data.frame)	A data.frame with 1431 rows and 22 columns
American Airlines.2019	list [4803 x 22] (S3: data.frame)	A data.frame with 4803 rows and 22 columns
Delta Air Lines.2019	list [2944 x 22] (S3: data.frame)	A data.frame with 2944 rows and 22 columns
Frontier Airlines.2019	list [1180 x 22] (S3: data.frame)	A data.frame with 1180 rows and 22 columns
Hawaiian Airlines.2019	list [225 x 22] (S3: data.frame)	A data.frame with 225 rows and 22 columns

`The 80 datasets were stored as grouped_data in R and in the bootstrap code block`

Using lapply(), bootstrap was done for each subset separately. As this was a 95% Confidence Interval, quartile range was 0.025 to 0.975, as this was a 2-sided (a flight can be sooner and also delayed, hence 2 sides)

The calculated confidence intervals were then graphed as error bars onto an average delay graph as error bars, with the center being the mean delay.

2. Standard Two-Way ANOVA & Welch's One-Way ANOVA

Standard Two-Way ANOVA (Fisher) was applied to compare whether Airports or Marketing Carriers have a bigger impact on delays. Originally I intended to use Aligned Rank Transform Two-Way ANOVA with the ARTool package. The reason being that it is non-parametric. However, in this dataset we do not have every combination of marketing carrier and airports thus the model could not be applied. Another limitation was that it required to be run overnight only to give error. Due to time constraint, ART ANOVA was abandoned and fallback of Standard ANOVA was used.

Although real world data is not normally distributed, the standard ANOVA test still applies as we are well past Central Limit Theorem (>30).

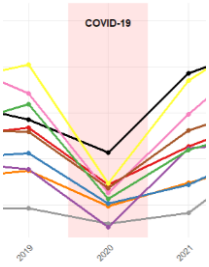
The Welch's One Way ANOVA was used for comparing the 10 marketing carriers for common airports only. The motivation for using the Welch variant specifically is same as the concerns during bootstrapping. Which is the non-normal nature of an on-time/delays dataset. Even if we account for Central Limit Theorem, it does not mean that the variances are going to be equal.

To justify using Welch's ANOVA was appropriate. I also ran Fligner-Killeen test as well, which gave a p-value < 0.05. Supporting the assumption that variances are not equal.

3. Holt-Winter's Time Series Forecast

This was used in the final section of the presentation to carry out a monthly forecast for 11 months between February and December of 2025. January was not modeled as it is already present in the dataset.

To carry out this test, monthly aggregate delays were calculated into new dataset “monthly_table” and converted to time-series object to be used with HoltWinters() function. The documentation for this function showed that the test can be fine tuned with extra arguments (alpha,beta,gamma). However, in this case I used the default HoltWinters model which is the additive one.



Drop in delay statistics for all carriers due to less traffic from Covid lockdown

The primary motive behind using HoltWinters for the time series forecast was it being a good for modeling the seasonal trends that are typical in monthly delay graphs. Yes, there is some limitation in this model where it would have been affected by the Covid-19 year in our dataset and throw off the accuracy, but HoltWinters() in R does not work with chronological breaks in the sample dataset it is trained on.

Analysis and Results

Results of this bootstrap test was used to graph the yearly historical trend with added error bars. As this test was done in a population dataset not a sample, the purpose of plotting these error bars was not to estimate some statistic in the population, but to check the stability of the performance put out by each marketing carrier.

```
                     marketing_carrier  avg_delay  ci_lower  ci_upper
Alaska Airlines.2018       Alaska Airlines 2018  6.709608  6.432326
7.016614
Allegiant Air.2018           Allegiant Air 2018 12.660360 12.102913
13.213829
American Airlines.2018   American Airlines 2018 11.793335 11.589783
11.999884
Delta Air Lines.2018       Delta Air Lines 2018 11.134220 10.089410
12.694918
Frontier Airlines.2018   Frontier Airlines 2018 19.922025 19.321505
20.620929
```

```
Hawaiian Airlines.2018   Hawaiian Airlines 2018 12.559178  9.492025
16.794720
JetBlue Airways.2018       JetBlue Airways 2018 16.388728 15.934613
16.889570
Southwest Airlines.2018 Southwest Airlines 2018 10.237688  9.999330
10.471111
Spirit Airlines.2018       Spirit Airlines 2018 12.215844 11.615157
12.815080
United Airlines.2018       United Airlines 2018 12.920595 12.472898
13.366007
Alaska Airlines.2019       Alaska Airlines 2019  7.728019  7.454646
8.006240
Allegiant Air.2019           Allegiant Air 2019 11.160497 10.604177
11.735911
American Airlines.2019   American Airlines 2019 11.986901 11.778537
12.216694
Delta Air Lines.2019       Delta Air Lines 2019 10.489242 10.019113
11.036612
Frontier Airlines.2019   Frontier Airlines 2019 15.342722 14.872253
15.875327
Hawaiian Airlines.2019   Hawaiian Airlines 2019  9.329103  8.497290
10.164900
JetBlue Airways.2019       JetBlue Airways 2019 18.163140 17.548519
18.798935
Southwest Airlines.2019 Southwest Airlines 2019  9.304480  9.036680
9.541355
Spirit Airlines.2019       Spirit Airlines 2019 11.800805 11.294236
12.324310
United Airlines.2019       United Airlines 2019 15.433773 14.879225
15.990132
Alaska Airlines.2020       Alaska Airlines 2020  4.130952  3.877300
4.427898
Allegiant Air.2020           Allegiant Air 2020  8.432900  8.001817
8.941140
American Airlines.2020   American Airlines 2020  6.084243  5.869962
6.284232
Delta Air Lines.2020       Delta Air Lines 2020  4.451601  4.217924
4.767646
Frontier Airlines.2020   Frontier Airlines 2020  4.993070  4.622093
5.373761
Hawaiian Airlines.2020   Hawaiian Airlines 2020  6.387785  4.611791
9.324319
JetBlue Airways.2020       JetBlue Airways 2020  6.061425  5.634647
6.490018
Southwest Airlines.2020 Southwest Airlines 2020  2.624048  2.517195
2.731769
Spirit Airlines.2020       Spirit Airlines 2020  5.316300  4.960884
5.669880
United Airlines.2020       United Airlines 2020  5.319360  4.850613
5.850767
Alaska Airlines.2021       Alaska Airlines 2021  6.943246  6.692487
7.226245
Allegiant Air.2021           Allegiant Air 2021 15.055115 14.452792
15.619552
American Airlines.2021   American Airlines 2021  9.732743  9.409024
10.161913
Delta Air Lines.2021       Delta Air Lines 2021  6.665690  6.311001
7.126829
Frontier Airlines.2021   Frontier Airlines 2021 11.639922 11.017003
12.339625
Hawaiian Airlines.2021   Hawaiian Airlines 2021  6.172551  5.272753
7.569826
JetBlue Airways.2021       JetBlue Airways 2021 15.851221 15.069804
16.699843
Southwest Airlines.2021 Southwest Airlines 2021 11.259790 10.906699
11.634827
Spirit Airlines.2021       Spirit Airlines 2021 11.432188 10.851468
12.061950
United Airlines.2021       United Airlines 2021 11.939994 11.302525
12.528471
Alaska Airlines.2022       Alaska Airlines 2022  8.918615  8.550965
9.257926
Allegiant Air.2022           Allegiant Air 2022 17.959690 17.256098
18.679872
American Airlines.2022   American Airlines 2022 12.685930 11.791288
14.446502
Delta Air Lines.2022       Delta Air Lines 2022 11.019789 10.525320
11.583669
Frontier Airlines.2022   Frontier Airlines 2022 17.813473 17.095602
18.597456
Hawaiian Airlines.2022   Hawaiian Airlines 2022 12.562325 10.964351
14.646750
JetBlue Airways.2022       JetBlue Airways 2022 22.542218 21.861304
23.244260
Southwest Airlines.2022 Southwest Airlines 2022 12.745583 12.481622
13.014405
Spirit Airlines.2022       Spirit Airlines 2022 15.254770 14.724017
15.783819
United Airlines.2022       United Airlines 2022 15.394163 14.694138
16.168009
Alaska Airlines.2023       Alaska Airlines 2023  8.334816  8.039000
8.664023
Allegiant Air.2023           Allegiant Air 2023 16.207563 15.383390
17.088609
American Airlines.2023   American Airlines 2023 13.397784 12.950243
13.923146
Delta Air Lines.2023       Delta Air Lines 2023 10.269653  9.936540
10.620608
Frontier Airlines.2023   Frontier Airlines 2023 19.385016 18.580068
20.164299
```

```
Hawaiian Airlines.2023   Hawaiian Airlines 2023 15.463795 13.782880
17.406643
JetBlue Airways.2023       JetBlue Airways 2023 23.829680 23.030661
24.721208
Southwest Airlines.2023 Southwest Airlines 2023 10.947614 10.665231
11.199254
Spirit Airlines.2023       Spirit Airlines 2023 18.768195 18.191164
19.454414
United Airlines.2023       United Airlines 2023 15.215166 14.708068
15.741915
Alaska Airlines.2024       Alaska Airlines 2024  8.883973  8.607088
9.170393
Allegiant Air.2024           Allegiant Air 2024 14.730015 14.022996
15.525129
American Airlines.2024   American Airlines 2024 16.036955 15.676068
16.408411
Delta Air Lines.2024       Delta Air Lines 2024 11.030172 10.693410
11.380041
Frontier Airlines.2024   Frontier Airlines 2024 17.735954 17.072010
18.359023
Hawaiian Airlines.2024   Hawaiian Airlines 2024  8.732118  7.874782
9.635299
JetBlue Airways.2024       JetBlue Airways 2024 18.159256 17.436117
18.908275
Southwest Airlines.2024 Southwest Airlines 2024 11.170940 10.883115
11.471693
Spirit Airlines.2024       Spirit Airlines 2024 14.178221 13.687078
14.677030
United Airlines.2024       United Airlines 2024 14.708745 14.159053
15.252059
Alaska Airlines.2025       Alaska Airlines 2025  8.023304  7.266671
8.788745
Allegiant Air.2025           Allegiant Air 2025 17.945841 15.374961
20.797957
American Airlines.2025   American Airlines 2025 14.974602 14.037808
15.959994
Delta Air Lines.2025       Delta Air Lines 2025 14.106514 13.202771
15.128827
Frontier Airlines.2025   Frontier Airlines 2025 18.133777 15.720754
20.496901
Hawaiian Airlines.2025   Hawaiian Airlines 2025 11.987321  9.842002
14.282654
JetBlue Airways.2025       JetBlue Airways 2025 15.350299 13.599976
17.267274
Southwest Airlines.2025 Southwest Airlines 2025  8.168333  7.651995
8.647525
Spirit Airlines.2025       Spirit Airlines 2025  9.261252  8.038065
10.551330
United Airlines.2025       United Airlines 2025 10.547196  9.899574
11.230006
```

As seen in the presentation, some airlines had much larger error bars for certain years due to lower sample sizes (example, Hawaiian Airlines in 2018). In contrast, Alaska and Southwest Airlines showed the most stability in their mean delay times for all years.

ANOVA Testing also showed significant limitation in this project, where the complexity of delay causes was illustrated in one of the result tables:

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Marketing Carrier	9	660,500	73,389	303.750	< 2e-16 ***
Airport	319	458,757	1,438	5.952	< 2e-16 ***
Residuals	110,188	26,622,500	242		

The huge number of residuals help illustrate an important characteristic of this dataset, where the **arr_delay** or mean delay statistics are affected by various factors, such as Weather, NOTAMs, Emergencies etc, which were not taken into account in order to keep the simplicity of this project for a single class. Still, for the sake of presentating something to the class, on basis of the 2-Way ANOVA F-Statistic, I justified ranking the carriers on common airports and generating confidence intervals on these means to setup the ranking:

```
Raw output for common_airport_ranking used to rank airlines
```

Note that in the graph, confidence intervals were calculated for a population statistic. The error bars were not to illustrate any uncertainty in the mean, but rather to show the stability of the carrier performance. This is not a conventional application for bootstrapping, when we already have the population statistic, however it does allow for us to estimate how each airline would perform if we 're-rolled' the scenario a 1000 times. In short, it was a crude way to use bootstrapping to some sort of forecasting.

Other methods were more straightforward, HoltWinters forecasting, despite being affected by the Covid-19 spike, did accurately predict the yearly trends in the 11 month period. Accurately showing peak delays in Summers (Caused by factors such as Holidays) and secondary peak in the winters (Caused by mainly Icing).

Conclusion:

To summarise, the project was able to arrive to a somewhat accurate conclusion, as per (Cirium,2024), where the rankings derived from the dataset are largely in-line with what an official report states. However, this is not surprising as we were dealing with a population dataset.

Establishing a true comparison of airline on time performance requires separating the target variable from many confounding factors, and the analysis demonstrated that arrival delay is a complex problem so attempting to reduce performance to a single average without controlling for route density or weather patterns yields an incomplete picture of operational efficiency.

The Two-Way ANOVA test provided critical insight into the structure of the dataset, and while both the carrier and the airport were found to be statistically significant causes of delay the large amount of leftover error indicates that most of the variance remains unexplained by these two factors alone. This high residual variance confirms that the dataset is statistically noisy because it is influenced by unobserved variables such as local weather events or air traffic control restrictions or mechanical logistics.

Despite the limitations of a streamlined statistical approach the models demonstrated strong external validity, and the seasonal trends identified by the Holt-Winters forecast such as the spike in summer and the drop in performance in December align closely with standard industry reports. This comparison suggests that while the absolute margin of error may vary the identified trends and relative performance rankings are robust. In future however, should employ a forecast model which is more resistant to anomalies like Covid-19 and is not dependent on chronology like HoltWinters

References:

1. Cirium. (2024). ***2024 monthly on-time performance report for airlines***. https://reports.cirium.com/2024-monthly-on-time-performance-report-for-airlines/full-view.html

2. Bureau of Transportation Statistics. (n.d.). *Airline on-time statistics and delay causes*. U.S. Department of Transportation. Retrieved December 7, 2025. https://www.transtats.bts.gov/ot_delay/ot_delaycause1.asp?qv52ynB=qn6n&20=E

Appendix:

Note: Plots used in presentation were more complex with ggtext used to load in .png images in the legend. This code will generate simpler plots that do not require those images to be on your computer.

```
setwd("C:/R/Project/")  * Change directory to where you have downloaded the airdata.csv



# Preface


# I have tried to keep the code to mostly base R, but I did have to resort to
# ggplot2, dplyr in certain cases to keep the code simple and make the graphs
# look better



# Rough Working Section (Please ignore this section)


# AIRLINE COLOURS
# 1. Colors
#airline_colors <- c(
 # "American Airlines"   = "#E41A1C",
 # "Delta Air Lines"     = "#377EB8",
 # "United Airlines"     = "#4DAF4A",
 # "Southwest Airlines"  = "#984EA3",
 # "Alaska Airlines"     = "#FF7F00",
 # "JetBlue Airways"     = "#FFFF33",
 # "Spirit Airlines"     = "#A65628",
 # "Frontier Airlines"   = "#F781BF",
 # "Hawaiian Airlines"   = "#999999",
 # "Allegiant Air"       = "#000000"  )
```

```
airdata_raw <- read.csv("Airline_Delay_Cause.csv", stringsAsFactors = FALSE)

test <- sum(airdata_raw$arr_flights, na.rm = TRUE)


dim(airdata_raw) # 398233 obs. of  21 variables

str(airdata_raw)

head(airdata_raw)

summary(airdata_raw)

airdata_raw <- na.omit(airdata_raw)

sd(airdata_raw$arr_del15)

sd(airdata_raw$arr_delay)

mean(airdata_raw$arr_delay)

mean(airdata_raw$arr_del15)









max_delay_row <- airdata_raw[which.max(airdata_raw$arr_delay), ]

max_delay_row


# Airline and Carrier Code Count


table_carrier <- table(airdata_raw$carrier)

table_carriername <- table(airdata_raw$carrier_name)

length(table_carrier)

length(table_carriername)







# Total Flights per carrier

flight_totals <- aggregate(arr_flights ~ carrier_name, data = airdata_raw, FUN = sum)

flight_totals_sorted <- flight_totals[order(flight_totals$arr_flights, decreasing = TRUE), ]

print(flight_totals_sorted)



#Airline Service Span

min_years <- aggregate(year ~ carrier_name, data = airdata_raw, FUN = min)
```

```
max_years <- aggregate(year ~ carrier_name, data = airdata_raw, FUN = max)



# End of Rough Working Section
#=============================================================================================================================



# Data Cleaning Steps.

# Main Issues:
# Compiling 51 reporting airlines to their respective marketing carrier
# Minor discrepancy in airport name vs airport code in some rows
# Omit negative delays. They appear to be an error in collection



airdata_raw$marketing_carrier <- airdata_raw$carrier_name



# American Airlines Group
AA_group<-c("American Airlines Inc.", "American Airlines Network", "Envoy Air", "Piedmont Airlines",
        "PSA Airlines Inc.", "US Airways Inc.", "America West Airlines Inc.", "American Eagle Airlines Inc.")

# Add to marketing_carrier
airdata_raw$marketing_carrier[airdata_raw$marketing_carrier %in% AA_group] <- "American Airlines"



# Delta Airlines
DAL_group <- c("Delta Air Lines Inc.", "Delta Air Lines Network", "Endeavor Air Inc.",
           "Northwest Airlines Inc.", "Comair Inc.", "Pinnacle Airlines Inc.",
           "Atlantic Southeast Airlines") # Major former regional partner
airdata_raw$marketing_carrier[airdata_raw$marketing_carrier %in% DAL_group] <- "Delta Air Lines"

# United Airlines
UA_group <- c("United Air Lines Inc.", "United Air Lines Network",
```

```
        "Continental Air Lines Inc.")
airdata_raw$marketing_carrier[airdata_raw$marketing_carrier %in% UA_group] <- "United Airlines"

# Alaska Airlines
AS_group <- c("Alaska Airlines Inc.", "Alaska Airlines Network",
        "Horizon Air", "Virgin America")
airdata_raw$marketing_carrier[airdata_raw$marketing_carrier %in% AS_group] <- "Alaska Airlines"

# Southwest
SW_group <- c("Southwest Airlines Co.", "Southwest Airlines",
        "AirTran Airways Corporation")
airdata_raw$marketing_carrier[airdata_raw$marketing_carrier %in% SW_group] <- "Southwest Airlines"


# Frontier
FA_group <- c("Frontier Airlines Inc.", "Frontier Airlines")
airdata_raw$marketing_carrier[airdata_raw$marketing_carrier %in% FA_group] <- "Frontier Airlines"

# Spirit
SA_group <- c("Spirit Air Lines", "Spirit Airlines")
airdata_raw$marketing_carrier[airdata_raw$marketing_carrier %in% SA_group] <- "Spirit Airlines"

# Hawaian
HA_group <- c("Hawaiian Airlines Inc.", "Hawaiian Airlines Network")
airdata_raw$marketing_carrier[airdata_raw$marketing_carrier %in% HA_group] <- "Hawaiian Airlines"



# NOTE: JetbBlue and Allegient is already sorted as required. Skipped.


# Vector of all marketing carriers

marketing_carrier_list <- c(
 "American Airlines",
 "Delta Air Lines",
 "United Airlines",
```

```
  "Southwest Airlines",
  "Alaska Airlines",
  "JetBlue Airways",
  "Spirit Airlines",
  "Frontier Airlines",
  "Hawaiian Airlines",
  "Allegiant Air"
)

# Add to dataset:

airdata_clean <- airdata_raw[airdata_raw$marketing_carrier %in% marketing_carrier_list,]
airdata_clean$marketing_carrier <- factor(airdata_clean$marketing_carrier)

# delete carrier_name
airdata_clean$carrier_name <- NULL



flight_totals <- aggregate(arr_flights ~ marketing_carrier, data = airdata_clean, FUN = sum)
flight_totals_sorted <- flight_totals[order(flight_totals$arr_flights, decreasing = TRUE), ]
print(flight_totals_sorted)

#marketing_carrier      arr_flights
#   American Airlines    31175293
#   Southwest Airlines   28248672
#   Delta Air Lines      26333133
#   United Airlines      13639313
#   Alaska Airlines      4965086
#   JetBlue Airways      4720711
#   Frontier Airlines    2161763
#   Spirit Airlines      1894520
#   Hawaiian Airlines    1437315
#   Allegiant Air        773860



# Matching airport names to airport code
```

```
unique_airport_pairs <- unique(airdata_clean[c("airport", "airport_name")])

length(unique_airport_pairs$airport) # 373


# Shifting to table
airport_code_counts <- table(unique_airport_pairs$airport)
airport_code_counts

# If count is greater than 1. Means a single code is corresponding to more than one name


problem_codes <- names(airport_code_counts[airport_code_counts > 1])
problem_pairs <- unique_airport_pairs[unique_airport_pairs$airport %in% problem_codes, ]



canonical_names <- unique_airport_pairs[!duplicated(unique_airport_pairs$airport), ]


airdata_clean$airport_name <- canonical_names$airport_name[ match(airdata_clean$airport, canonical_names$airport) ]

# Check code worked
airport_code_test <- unique(airdata_clean[c("airport", "airport_name")])
any(duplicated(airport_code_test$airport))

# Negative Delays

negative_causes <- sum(
airdata_clean$arr_delay <0,
airdata_clean$arr_del15 <0,
airdata_clean$carrier_delay < 0,
airdata_clean$weather_delay < 0,
airdata_clean$nas_delay < 0,
airdata_clean$security_delay < 0,
airdata_clean$late_aircraft_delay < 0,
```

```
na.rm = TRUE)

print(negative_causes)

# 2 Rows where delays are negative. Due to their insignificance, we will remove

airdata_clean <- subset(
airdata_clean,
carrier_delay >= 0 &
weather_delay >= 0 &
nas_delay >= 0 &
security_delay >= 0 &
late_aircraft_delay >= 0 &
arr_delay >= 0 &
arr_del15 >= 0)




summary(airdata_clean$marketing_carrier)
length(airdata_clean$marketing_carrier)



# We are only working for the years 2018-2025 as this is the time period
# where all marketing carriers have flight data (for this dataset)

airdata <- subset(airdata_clean, year >= 2018)

#_____________________________________________________________________________________________________



# Part 1: Historical Trend Analysis
```

```
library(ggplot2)

# Usage: Graphing


library(ggtext)

# Usage: experimenting with adding airline logos as pictures to the graphs


# I planned to use dplyr but project was doable with Base R only




# Aggregate delay for Marketing Carriers

agg_delay <- aggregate(arr_delay ~ marketing_carrier + year, data = airdata, FUN = sum)


# Total Flights for each carrier

agg_flights <- aggregate(arr_flights ~ marketing_carrier + year, data = airdata, FUN = sum, na.rm = TRUE)


# Merge tables

annual_trends <- merge(agg_delay, agg_flights, by = c("marketing_carrier", "year"))


# Calculate the average delay per flight

annual_trends$avg_delay <- annual_trends$arr_delay / annual_trends$arr_flights







# Aircraft Logos (ggtext) and Colours


# 1. Colors for Graph

airline_colors <- c(

 "American Airlines"   = "#E41A1C", # Red

 "Delta Air Lines"     = "#377EB8", # Blue

 "United Airlines"     = "#4DAF4A", # Green

 "Southwest Airlines"  = "#984EA3", # Purple

 "Alaska Airlines"     = "#FF7F00", # Orange

 "JetBlue Airways"     = "#FFFF33", # Yellow

 "Spirit Airlines"     = "#A65628", # Brown

 "Frontier Airlines"   = "#F781BF", # Pink

 "Hawaiian Airlines"   = "#999999", # Grey

 "Allegiant Air"       = "#000000"  # Black
```

```
)



# Plotting The Graph

trend_plot <- ggplot(annual_trends, aes(x = year, y = avg_delay, color = marketing_carrier)) +

 # --- COVID-19 Highlight ---
 annotate("rect", xmin = 2019.5, xmax = 2020.5, ymin = -Inf, ymax = Inf,
          alpha = 0.1, fill = "red") +
 annotate("text", x = 2020, y = max(annual_trends$avg_delay),
          label = "COVID-19", vjust = 1.5, size = 3.5, fontface = "bold") +

 # Lines and Points
 geom_line(linewidth = 1) +
 geom_point(size = 2) +

 # Colors
 scale_color_manual(values = airline_colors) +

 # Labels
 labs(
  title = "Average Arrival Delay per Flight (2018-2025)",
  subtitle = "Historical Performance of 10 Major U.S. Marketing Carriers",
  x = "Year",
  y = "Average Delay per Flight (Minutes)"
 ) +

 # Theme and formatting
 scale_x_continuous(breaks = 2018:2025) +
 theme_minimal() +
 theme(
  legend.position = "bottom",
  axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1),
  panel.grid.minor = element_blank()
 )
```

```
print(trend_plot)
```

```
#=============================================================================================================================
# Bootstrapping Confidence Interval for Population Trend

set.seed(7100) # same result on diff. machines

bootstrap_function <- function(delays, n_boot = 1000) {
 # ^this creates a function
delays <- na.omit(delays)
n <- length(delays)


boot_means <- replicate(n_boot, mean(sample(delays, size = n, replace = TRUE)))

ci <- quantile(boot_means, probs = c(0.025, 0.975))

return(c(avg_delay = mean(delays), ci_lower = ci[1], ci_upper = ci[2]))
}

#ci[1] is lower
#ci[2] is upper


# New variable in dataset without 0s

airdata$delay_per_flight_row <- ifelse(
 airdata$arr_flights == 0, 0,
 airdata$arr_delay / airdata$arr_flights
)

# dividing by zero will cause error which is why we need this

# Split data
grouped_data <- split(airdata, list(airdata$marketing_carrier, airdata$year), drop = TRUE)
```

```
# creates 80 different datasets

# we are splliting airdata by each marketing carrrier and each year

# for example, Delta.2019, United.2020 etc.



# Run bootstrap

bootstrap_list <- lapply(grouped_data, function(df_subset) {


# df_subset is stand in for the 80 subsetted datasets we created with split


stats <- bootstrap_function(df_subset$delay_per_flight_row)


data.frame(

 marketing_carrier = df_subset$marketing_carrier[1],

 # checks 1st row for name of carrier

 year           = df_subset$year[1],

 # checks 1st row for the year

 avg_delay       = stats["avg_delay"],

 ci_lower        = stats["ci_lower.2.5%"],

 ci_upper        = stats["ci_upper.97.5%"],

 row.names       = NULL

 )

})


# Result

bootstrap_results <- do.call(rbind, bootstrap_list)


# Run to check

bootstrap_results


#Adding the CIs to the line graph


library(ggplot2)


pd <- position_dodge(0.5)


boot_plot <- ggplot(bootstrap_results, aes(x = year, y = avg_delay, color = marketing_carrier)) +
```

```
  # covid-19 highlight
  annotate("rect", xmin = 2019.5, xmax = 2020.5, ymin = -Inf, ymax = Inf,
           alpha = 0.1, fill = "red") +
  annotate("text", x = 2020, y = max(bootstrap_results$ci_upper, na.rm=TRUE),
           label = "COVID-19", vjust = 1.5, size = 3.5, fontface = "bold") +

  geom_line(position = pd, linewidth = 0.8, alpha = 0.6) +
  geom_point(position = pd, size = 2) +
  geom_errorbar(aes(ymin = ci_lower, ymax = ci_upper), position = pd, width = 0.4, linewidth = 0.8) +

  scale_color_manual(values = airline_colors) +
  scale_x_continuous(breaks = 2018:2025)
  labs(
    title = "Bootstrapped 95% CI for Delay per Flight (2018-2025)",
    subtitle = "Comparison of Sample Mean vs. Population Estimate Uncertainty",
    x = "Year",
    y = "Average Delay per Flight (Minutes)"
  ) +

  theme_minimal() +
  guides(color = guide_legend(nrow = 2, byrow = TRUE)) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1),
    panel.grid.minor = element_blank()
  )

print(boot_plot)
```

```
#=============================================================================================================================================
=
```

```
# Airport Traffic Heatmap



# Find Top 20 Busiest Airports by Total Flights

airport_vols <- aggregate(arr_flights ~ airport, data = airdata, FUN = sum)

# Note: airport_vols is Airport Volumes

top_20_codes <- airport_vols[order(-airport_vols$arr_flights), "airport"][1:20]


#[1:20] cuts off for top 20 instead of 316 airports


# Note: Code refers to the airport codes here.

# In aviation we had 3 letter identifiers for airports.

# For example ATL is Hartsfield Jackson airport in Atlanta


# Filter dataset to keep only rows with these airports

airdata_heatmap <- subset(airdata, airport %in% top_20_codes)



# Omit zeros to prevent division error


airdata_heatmap$delay_per_flight <- ifelse(

airdata_heatmap$arr_flights == 0, 0,

airdata_heatmap$arr_delay / airdata_heatmap$arr_flights

)


# Heatmap


# Aggregate mean delay for each Airline-Airport pair

heatmap_agg <- aggregate(

 delay_per_flight ~ marketing_carrier + airport,

 data = airdata_heatmap,

 FUN = mean

)


# Create Heatmap

heat_plot <- ggplot(heatmap_agg, aes(x = airport, y = marketing_carrier, fill = delay_per_flight)) +

 geom_tile(color = "white") +
```

```
# Colour Theme
scale_fill_gradientn(colors = c("white", "#FFC629", "black")) +

labs(
title = "Heatmap: Average Delay by Airline and Airport (Top 20 Airports)",
x = "Airport Code",
y = "Marketing Carrier",
fill = "Avg Delay (min)"
) +

theme_minimal() +
theme(
panel.grid = element_blank(),
axis.text.x = element_text(angle = 45, hjust = 1)
)

print(heat_plot)



# Conclusion:
# Jetblue and Frontier are consisently poor performers across all airports
# however,
# EWR (Newark Liberty) is consistently causing high delay across all carriers
# Graph supports theory that airports are a bigger player in delay
# lets confirm with ANOVA test




# The ANOVA test is not being run on the heatmap subset.
# Entire dataset is being considered



# Formula: Delay is explained by Carrier + Airport
two_way_ANOVA <- aov(delay_per_flight_row ~ marketing_carrier + airport, data = airdata)

# Print the summary to see the F-statistics side-by-side
summary(two_way_ANOVA)
```

two_way_ANOVA

My Conclusion:

Marketing Carrier have a larger impact on delay cause than the marketing carrier

Comparing airlines without taking into account the differnce of routes is unfair

#===

Airport adjusted on-time performance comparison

Setting up new dataset

airport_carrier_counts <- aggregate(

 marketing_carrier ~ airport,

 data = airdata,

 FUN = function(x) length(unique(x))

)

x is marketing_carrier in the above code

#length(unique(x)) will be equal to 10

airport_carrier_counts

shared_airports <- airport_carrier_counts$airport[airport_carrier_counts$marketing_carrier == 10]

Filters out airport_carrier_counts where marketing_carrier<10

print(shared_airports)

Subset airdata for shared airports only

airdata_common_airports <- subset(airdata, airport %in% shared_airports)

nrow(airdata_common_airports) # 7238 observations for common airports

Check to see all marketing carriers have enough flights to do analysis on for these airports

```
marketing_carrier_counts <- table(airdata_common_airports$marketing_carrier)



# Average delay per flight for each row


#ANOVA on Common Airports






# Justificaiton for using Welch
# CLT applies to a large dataset
# That does not include equal variance
variance_test <- fligner.test(delay_per_flight_row ~ marketing_carrier, data = airdata)
print(variance_test) # p-value is 2.2e-16

# Conclusion: Using Welch ANOVA For one way tests
#  Create the Performance Metric (Row-Level)
#    We need to calculate the average delay per flight for each row
#    in this specific subset before we can run the test.



airdata_common_airports$delay_per_flight <- ifelse(
 airdata_common_airports$arr_flights == 0, 0,
 airdata_common_airports$arr_delay/
 airdata_common_airports$arr_flights
)



anova_common_airports <- oneway.test(
 delay_per_flight ~ marketing_carrier,
 data = airdata_common_airports,
 var.equal = FALSE
)


print(anova_common_airports)
```

```
# p value is less than 0.05, proceed with ranking:
common_airport_ranking <- aggregate(
  delay_per_flight ~ marketing_carrier,
  data = airdata_common_airports,
  FUN = mean
)

# Sort by lowest delay (The Winner)
common_airport_ranking <- common_airport_ranking[order(common_airport_ranking$delay_per_flight), ]



print(common_airport_ranking)




# --- Step 1: Prepare the Statistics ---

# We already have the Mean in 'common_airport_ranking'

#  Calculate Standard Deviation
common_airport_sd <- aggregate(
  delay_per_flight ~ marketing_carrier,
  data = airdata_common_airports,
  FUN = sd
)

# Calculate Count
common_airport_n <- aggregate(
  delay_per_flight ~ marketing_carrier,
  data = airdata_common_airports,
  FUN = length
)



# Merge  Mean and SD
# Since both have a column named "delay_per_flight", R will rename them:
# delay_per_flight.x and delay_per_flight.y by default
```

```
plot_data <- merge(common_airport_ranking, common_airport_sd, by = "marketing_carrier")

# Merge the result with N
# The new column from common_airport_n will stay as "delay_per_flight"
plot_data <- merge(plot_data, common_airport_n, by = "marketing_carrier")


# --- Step 3: Calculate Confidence Intervals ---

# We now reference the columns based on how R merged them:
# Mean = delay_per_flight.x
# SD   = delay_per_flight.y
# N    = delay_per_flight

# Standard Error (SE = sd / sqrt(n))
plot_data$se <- plot_data$delay_per_flight.y / sqrt(plot_data$delay_per_flight)

# Margin of Error (95% Confidence)
plot_data$margin <- qt(0.975, df = plot_data$delay_per_flight - 1) * plot_data$se

# Calculate Bounds using the Mean (.x)
plot_data$lower_ci <- plot_data$delay_per_flight.x - plot_data$margin
plot_data$upper_ci <- plot_data$delay_per_flight.x + plot_data$margin


# --- Step 4: Create the Plot ---
ksu_gold <- "#FFC629"
ksu_black <- "#000000"

# Note: We use delay_per_flight.x (the Mean) for the x-axis and reordering
ggplot(plot_data, aes(x = reorder(marketing_carrier, -delay_per_flight.x), y = delay_per_flight.x)) +

  # Error Bars
  geom_errorbar(aes(ymin = lower_ci, ymax = upper_ci), width = 0.2, color = ksu_black, size = 0.8) +

  # The Points
  geom_point(size = 4, color = ksu_gold) +
```

```
 # Formatting
 coord_flip() +
 labs(
  title = "Airline Delay Ranking (Common Airports)",
  subtitle = "Mean Delay per Flight with 95% Confidence Intervals",
  x = "Airline",
  y = "Average Delay (Minutes)"
 ) +

 theme_minimal() +
 theme(
  plot.title = element_text(face = "bold", size = 16, color = ksu_black),
  plot.subtitle = element_text(face = "italic", size = 11, color = "gray30"),
  axis.text = element_text(color = ksu_black, size = 10),
  axis.title = element_text(face = "bold", color = ksu_black),
  panel.grid.major.y = element_blank()
 )



# Part 4 Time Series Forecast (Holt-Winters)

?HoltWinters()
monthly_data <- aggregate(
cbind(arr_delay, arr_flights) ~ marketing_carrier + year + month,
data = airdata, # Assumes 'airdata' is your filtered 2018-2025 dataset
FUN = sum
)

# avergae delay variable

monthly_data$avg_delay <- monthly_data$arr_delay / monthly_data$arr_flights

# Sort by date
monthly_data <- monthly_data[order(monthly_data$marketing_carrier, monthly_data$year, monthly_data$month), ]



# Forecast Function
```

```
predict_airline_performance <- function(airline) {



data_sub <- subset(monthly_data, marketing_carrier == airline)



ts_data <- ts(data_sub$avg_delay, start = c(2018, 1), frequency = 12)

# Apply HoltWinters

model <- HoltWinters(ts_data)

pred  <- predict(model, n.ahead = 11, prediction.interval = TRUE, level = 0.95)



# Print results to table (data frame)

  return(data.frame(
    Airline = airline,
    Month = c("Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
    Prediction = round(as.numeric(pred[, "fit"]), 2),
    Lower_CI   = round(as.numeric(pred[, "lwr"]), 2),
    Upper_CI   = round(as.numeric(pred[, "upr"]), 2)
  ))
}




# Run the function for all 10 airlines

all_airlines <- unique(monthly_data$marketing_carrier)

final_forecast <- do.call(rbind, lapply(all_airlines, predict_airline_performance))

# Print the table

print(head(final_forecast, 11))

ignore <- unique(final_forecast$Airline)

# Delta Airlines TSF

plot_data_delta <- subset(final_forecast, Airline == "Delta Air Lines")
```

```
# Add a numeric index for plotting order
plot_data_delta$Month_Index <- 1:11

# Graph

ggplot(plot_data_delta, aes(x = Month_Index, y = Prediction)) +
geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
geom_line(color = "#FFC629", linewidth = 1) +
geom_point(color = "black", size = 3) +
scale_x_continuous(breaks = 1:11, labels = plot_data_delta$Month) +
labs(title = "Delta Air Lines: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
theme_minimal()



# United Airlines TSF
plot_data_united <- subset(final_forecast, Airline == "United Airlines")
plot_data_united$Month_Index <- 1:11



# Linegraph
ggplot(plot_data_united, aes(x = Month_Index, y = Prediction)) +
 geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
 geom_line(color = "#FFC629", linewidth = 1) +
 geom_point(color = "black", size = 3) +
 scale_x_continuous(breaks = 1:11, labels = plot_data_united$Month) +
 labs(title = "United Airlines: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
 theme_minimal()



# Alaska Airlines TSF
plot_data_alaska <- subset(final_forecast, Airline == "Alaska Airlines")
plot_data_alaska$Month_Index <- 1:11

# Graph
ggplot(plot_data_alaska,aes(x = Month_Index, y = Prediction)) +
 geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
 geom_line(color = "#FFC629", linewidth = 1) +
 geom_point(color = "black", size = 3) +
```

```
  scale_x_continuous(breaks = 1:11, labels = plot_data_alaska$Month) +
  labs(title = "Alaska Airlines: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
  theme_minimal()

# Allegiant Air TSF
plot_data_allegiant <- subset(final_forecast, Airline == "Allegiant Air")
plot_data_allegiant$Month_Index <- 1:11

ggplot(plot_data_allegiant, aes(x = Month_Index, y = Prediction)) +
  geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
  geom_line(color = "#FFC629", linewidth = 1) +
  geom_point(color = "black", size = 3) +
  scale_x_continuous(breaks = 1:11, labels = plot_data_allegiant$Month) +
  labs(title = "Allegiant Air: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
  theme_minimal()


# American Airlines TSF
plot_data_american <- subset(final_forecast, Airline == "American Airlines")
plot_data_american$Month_Index <- 1:11

ggplot(plot_data_american, aes(x = Month_Index, y = Prediction)) +
  geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
  geom_line(color = "#FFC629", linewidth = 1) +
  geom_point(color = "black", size = 3) +
  scale_x_continuous(breaks = 1:11, labels = plot_data_american$Month) +
  labs(title = "American Airlines: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
  theme_minimal()


# Frontier Airlines TSF
plot_data_frontier <- subset(final_forecast, Airline == "Frontier Airlines")
plot_data_frontier$Month_Index <- 1:11

ggplot(plot_data_frontier, aes(x = Month_Index, y = Prediction)) +
  geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
  geom_line(color = "#FFC629", linewidth = 1) +
  geom_point(color = "black", size = 3) +
```

```
  scale_x_continuous(breaks = 1:11, labels = plot_data_frontier$Month) +
  labs(title = "Frontier Airlines: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
  theme_minimal()


# Hawaiian Airlines TSF
plot_data_hawaiian <- subset(final_forecast, Airline == "Hawaiian Airlines")
plot_data_hawaiian$Month_Index <- 1:11

ggplot(plot_data_hawaiian, aes(x = Month_Index, y = Prediction)) +
  geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
  geom_line(color = "#FFC629", linewidth = 1) +
  geom_point(color = "black", size = 3) +
  scale_x_continuous(breaks = 1:11, labels = plot_data_hawaiian$Month) +
  labs(title = "Hawaiian Airlines: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
  theme_minimal()


# JetBlue Airways TSF
plot_data_jetblue <- subset(final_forecast, Airline == "JetBlue Airways")
plot_data_jetblue$Month_Index <- 1:11

ggplot(plot_data_jetblue, aes(x = Month_Index, y = Prediction)) +
  geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
  geom_line(color = "#FFC629", linewidth = 1) +
  geom_point(color = "black", size = 3) +
  scale_x_continuous(breaks = 1:11, labels = plot_data_jetblue$Month) +
  labs(title = "JetBlue Airways: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
  theme_minimal()


# Southwest Airlines TSF
plot_data_southwest <- subset(final_forecast, Airline == "Southwest Airlines")
plot_data_southwest$Month_Index <- 1:11

ggplot(plot_data_southwest, aes(x = Month_Index, y = Prediction)) +
  geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
  geom_line(color = "#FFC629", linewidth = 1) +
```

```
  geom_point(color = "black", size = 3) +
  scale_x_continuous(breaks = 1:11, labels = plot_data_southwest$Month) +
  labs(title = "Southwest Airlines: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
  theme_minimal()



# Spirit Airlines TSF
plot_data_spirit <- subset(final_forecast, Airline == "Spirit Airlines")
plot_data_spirit$Month_Index <- 1:11

ggplot(plot_data_spirit, aes(x = Month_Index, y = Prediction)) +
  geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
  geom_line(color = "#FFC629", linewidth = 1) +
  geom_point(color = "black", size = 3) +
  scale_x_continuous(breaks = 1:11, labels = plot_data_spirit$Month) +
  labs(title = "Spirit Airlines: 2025 Forecast", y = "Predicted Delay (min)", x = "Month") +
  theme_minimal()




#=============================================================================================



final_forecast$Month_Index <- rep(1:11, times = 10)

# Grid Plot
ggplot(final_forecast, aes(x = Month_Index, y = Prediction)) +



  geom_ribbon(aes(ymin = Lower_CI, ymax = Upper_CI), alpha = 0.2, fill = "#FFC629") +
  geom_line(color = "#FFC629", linewidth = 1) +
  geom_point(color = "black", size = 1.5) + # Slightly smaller dots for the small graphs

# Facet Wrap for 2 row of 5 each
  facet_wrap(~ Airline, nrow = 2, ncol = 5, scales = "free_y") +

  # Formatting
```

```
labs(x = NULL, y = NULL) +


theme_minimal() +
theme(
  # Remove axis titles and text (ticks) to reduce clutter
  axis.title = element_blank(),
  axis.text = element_blank(),

  # Make the grid lines subtle
  panel.grid.major = element_line(color = "grey90"),
  panel.grid.minor = element_blank(),

  # Make the Airline Names (Strip Text) bold and readable
  strip.text = element_text(face = "bold", size = 10, color = "black")
)
```